Cross Fusion Net: A Fast Semantic Segmentation Network for Small-Scale Semantic Information Capturing in Aerial Scenes

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***Abstract*— Capturing accurate multiscale semantic information from the images is of great importance for high-quality semantic segmentation. Over the past years, a large number of methods attempt to improve the multiscale information capturing ability of the networks via various means. However, these methods always suffer unsatisfactory efficiency (e.g., speed or accuracy) on the images that include a large number of small-scale objects, for example, aerial images. In this article, we propose a new network named cross fusion net (CF-Net) for fast and effective extraction of the multiscale semantic information, especially for small-scale semantic information. In particular, the proposed CF-Net can capture more accurate small-scale semantic information from two aspects. On the one hand, we develop a channel attention refinement block to select the informative features. On the other hand, we propose a cross fusion block to enlarge the receptive field of the low-level feature maps. As a result, the network can encode more accurate semantic information from the small-scale objects, and the segmentation accuracy of the small-scale objects is improved accordingly. We have compared the proposed CF-Net with several state-of-the-art semantic segmentation methods on two popular aerial image segmentation data sets. Experimental results reveal that the average F_1 score gain brought by our CF-Net is about 0.43% and the F_1 score gain of the small-scale objects (*e.g.*, cars) is about 2.61%. In addition, our CF-Net has the fastest inference speed, which proves its superiority in the aerial scenes. Our code will be released at: https://github.com/pcl111/CF-Net.**

***Index Terms*— Aerial image, deep learning, multiscale, semantic segmentation, small scale.**

I. Introduction

IN THE remote sensing field, semantic segmentation technology is of great importance, which can be widely used in environmental monitoring [1], [2], agriculture [3], [4], forestry [5], [6], urban planning [7], [8], and so on. Traditional methods [9], [10] address this problem by designing features and classifiers manually and exhibit unsatisfactory performance due to their inferior capabilities of feature extraction and data fitting. Over the past years, benefiting from the success of deep learning [11]–[15], an increasing number of deep learning-based methods [16]–[18] have been proposed to address this problem and surpassed traditional methods with a large margin. In particular, most deep learning-based methods tend to increase the segmentation accuracy via addressing an ill-posed problem, i.e., multiscale semantic information capturing.

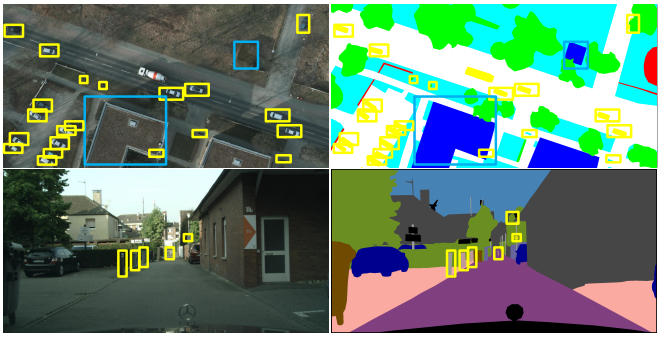

Fig. 1. Blue boxes display the importance of multiscale semantic information capturing. The yellow boxes illustrate the difference between (Top) aerial image and (Bottom) ground-based image. Clearly, the aerial image has much more small-scale objects.

The importance of multiscale semantic information capturing on the semantic segmentation task is shown in the blue boxes of Fig. 1. To achieve high-quality segmentation, a network should classify the classes of pixels from multiple scales rather than a fixed scale. For this purpose, there are mainly three kinds of deep learning-based methods (i.e., pyramid structure, multicolumn structure, and encoder–decoder network) proposed in recent years. However, these methods applied to ground-based images cannot achieve high speed and accuracy simultaneously when handling aerial images. Specifically, compared with the ground-based images, the aerial images have a larger field of view (FOV) under the same resolution. As a result, the aerial images often have more small-scale objects (e.g., the cars and buildings), as shown in the yellow boxes of Fig. 1. Therefore, the semantic segmentation network applied to aerial scenes should possess strong ability to handle small-scale objects. However, previous methods typically cannot acquire accurate small-scale semantic

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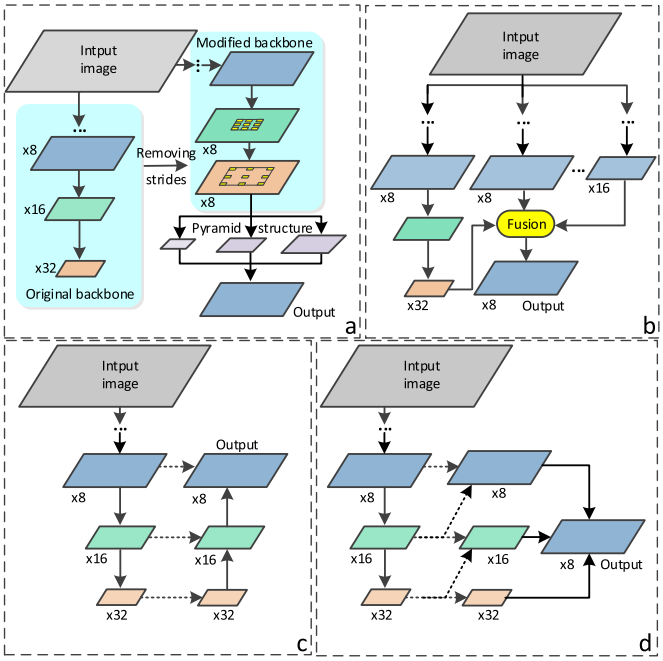

Fig. 2. Structures for multiscale semantic information capturing. (a) Pyramid structure. (b) Multicolumn network. (c) Encoder–decoder network. (d) Our proposed CF-Net.

information at a fast speed, which hampers their application to the aerial scenes of large size. In the following, the limitations of the existing methods will be elaborated.

For the first solution, i.e., the pyramid structure as shown in Fig. 2(a), it consists of multiple branches and each branch can extract semantic information from a corresponding scale. To ensure that the small object can be detected, recent methods based on the pyramid structure always discard some strides in backbone networks, which can increase the input resolution of the pyramid structure significantly. However, this modification will bring a large amount of computation, which hampers its practical application. The second method is the multicolumn network, as shown in Fig. 2(b). It uses multiple branches whose output resolutions are different to capture semantic information from different scales. To save the inference time, previous real-time methods always adopt shallow branches to generate the feature maps whose resolutions are large, leading to their inferior semantic information capturing ability on small-scale objects. Therefore, the multicolumn network can hardly attain the accuracy requirement. The last method is the encoder–decoder network, as shown in Fig. 2(c). It utilizes the feature maps from different blocks of the backbone network to capture multiscale semantic information. However, previous works always adopt simple backbone networks to achieve high inference speed. If we use feature maps from shallow blocks of the backbone network to capture semantic information, the encoder–decoder-based method will suffer from a similar problem as the multicolumn network. As a result, this method cannot achieve the accuracy requirements in the aerial image semantic segmentation field either.

In this article, we pay attention to design a lightweight yet effective network to perform fast semantic segmentation for aerial scenes, as shown in Fig. 2(d). Similar to the encoder–decoder structure, the feature maps from different stages of the backbone network are used to capture multiscale semantic information, and their semantic information capturing ability can be enhanced from two aspects. In particular, we first use a popular deep learning technique (i.e., attention mechanism) to assign weights for the channels of the feature maps. With this operation, the feature maps can have more informative features, leading to their enhanced semantic information encoding ability. Besides, we improve the receptive field of feature maps from the shallow stages (i.e., low-level feature maps used to capture small-scale semantic information) by fusing them with feature maps from the deep stages (i.e., high-level feature maps). With the guidance information provided by the high-level feature maps, the low-level ones will have similar information capturing ability as the high-level ones. As a result, the network can extract more accurate semantic information from small-scale objects. To evaluate the superiority of the proposed cross fusion net (CF-Net), we use two famous image classification networks (e.g., ResNet-18 and VGG-16) as the backbones to perform fast semantic segmentation. We compare the CF-Net with several popular aerial image semantic segmentation methods on two aerial image segmentation benchmarks, and the results illustrate that the CF-Net can obtain better segmentation performance on the small-scale objects with a faster inference speed.

The main contribution of this work is described as follows. We first analyze the importance of the small-scale semantic information capturing for the semantic segmentation of aerial scenes. Accordingly, we propose a fast semantic segmentation network with a lightweight backbone named CF-Net for multiscale semantic information capturing in aerial images, which can refine the hierarchical feature maps via two blocks, i.e., a channel attention refinement block to select informative features and a cross fusion block to increase the receptive field of low-level feature maps. With these two blocks, the accuracy of semantic information from different scales (e.g., especially from the small scale) can be increased significantly, leading to a more efficient network for aerial scene segmentation.

II. Related Work

In this section, several representational studies related to our work will be presented. In particular, some semantic segmentation methods used to capture the multiscale semantic segmentation and applied in aerial scenes will be introduced.

A. Pyramid Structure

The pyramid structure consists of two main forms. The first kind can capture multiscale semantic information by changing the input resolution of the feature maps. For instance, PSPNet [19] generates feature maps of different resolutions directly via the global average pooling operation. SSPP [20] uses the pooling operation with strides to change the size of the feature maps. The second kind extracts multiscale semantic information by changing the rates of convolution layers. For example, DeepLab [21], [22] uses multiply convolution layers whose rates are different to capture multiscale semantic information. DensASPP [23] improves the large-scale semantic

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information capturing ability of DeepLab by introducing dense connection to augment the receptive field. However, these methods can hardly be applied to practical remote sensing applications due to their inferior speed caused by the following two shortages. The first is the computation burden brought by the pyramid structure. In detail, its input feature map is the deepest feature map of the backbone network that has more channels. To extract accurate semantic information from this feature map, the pyramid structure usually needs to possess a complex structure, leading to a low inference speed. The second shortage is caused by the change of the backbone network. In particular, the resolution of the deepest feature map of the backbone network is too small to satisfy the small-scale semantic information capturing requirement [24]. To ensure the input resolution of the pyramid structure, previous methods modify the baseline network by discarding certain strides and introducing atrous convolutions into the networks [23], [24]. With this modification, the output resolution of the baseline network can be increased significantly. Nevertheless, after removing the strides in deep blocks, it will bring a large amount of computation.

B. Multicolumn Structure

The multicolumn networks can capture multiscale semantic information by introducing several branches whose output resolutions are different. For instance, ICNet [25] uses two spatial paths to capture small-scale objects and fuses different scale output feature maps via image cascade. In [26], branches that possess large output resolutions are used to generate a scale score map to assign weights for low-resolution branches. BiseNet [27] employs a lightweight model to extract the semantic information from the small-scale objects. Fast-SCNN [28] reduces computation by sharing the shallow layers' parameters of different branches. Nevertheless, the contradiction between the network depth and the information capturing ability is still a challenge for the practical application of the multicolumn networks. In particular, to improve their semantic information capturing ability, the branches of a multicolumn network should have a deep enough network structure, which can ensure its large receptive field and the ability to extract more abstract information. Nevertheless, increasing the network depth will sacrifice the inference speed, especially for the branches whose resolution is large. Previous fast semantic segmentation methods based on the multicolumn network always adopt pretty shallow branches to generate the feature maps whose resolutions are large, leading to their inferior semantic information capturing ability on small-scale objects. Consequently, the multicolumn network can hardly attain the accuracy requirement of aerial image segmentation.

C. Encoder–Decoder Structure

The encoder–decoder structure can acquire multiscale semantic information by fusing hierarchical feature maps whose resolutions are different. In particular, the fully convolutional network (FCN) [16] incorporates the low- and high-level feature maps via proposing a skip structure. U-net [29] can achieve high fusion accuracy by developing a high-efficiency skip connection network structure. SegNet [30] can record saved pool indices in the encoding process and uses these indices to supervise the decoding process. RefineNet [31] introduces a large number of refinement blocks to enhance the semantic information capturing the ability of hierarchical feature maps, whereas LW-RefineNet [32] optimizes RefineNet by removing redundancy refinement blocks and can attain real-time performance. Although these encoder–decoder structures are able to capture multiscale semantic information in a fast way, the encoder–decoder network also suffers from the similar problem as the multicolumn network. In particular, to achieve a high inference speed, the backbone network should be as simple as possible and hence has a shallow structure. If we utilize the middle feature maps to obtain the small-scale semantic information, it will be inaccurate due to the limited receptive field. As a result, the encoder–decoder network cannot attain high segmentation accuracy on small-scale objects either.

D. Aerial Image Semantic Segmentation

Over the past years, deep learning has been applied to the aerial image semantic segmentation field and expressed powerful segmentation ability. For instance, ERN [33] eliminates the influence of the semantic ambiguity by introducing the spatial boundary context, which can address the high interclass similarity problem and eliminate the interference from ubiquitous shadows. To automatically carry out lane-marking segmentation in aerial imagery, HFSA-Unet [18] explores both the global mean and the inner product among different channels to adaptively rescale intermediate features. RA-FCN [34] uses a spatial relation module and a channel relation module to learn and reason about global relationships between any two spatial positions or feature maps. CGFDN [35] improves segmentation accuracy via extracting more contextual information from the aerial image. Although these methods have achieved great segmentation performance, they can hardly be applied to practical aerial scenes due to their inferior speed. Different from these methods, our proposed CF-Net can achieve high-quality segmentation performance at a fast speed.

III. Proposed Method

The whole process to extract multiscale semantic information from aerial images can be decomposed into three steps (i.e., information encoding, refinement, and fusion). In this section, we first introduce the specific blocks to conduct these three steps and then fuse them to construct the proposed CF-Net.

A. Information Encoding

To encode accurate semantic information from the input images, a network should possess a large enough receptive field to ensure the information capturing ability of the network. To this end, we adopt two popular image classification networks (i.e., VGG-16 [36] and ResNet-18 [11]) as our backbone networks to encode the semantic information. Next, we will introduce their specific implementations.

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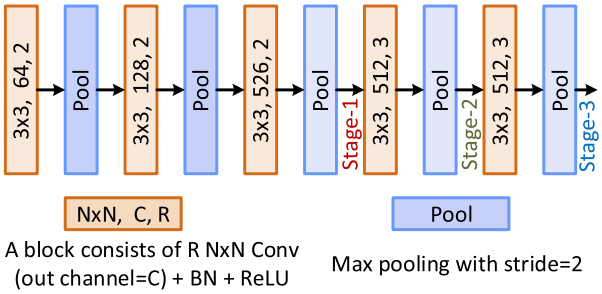

Fig. 3. Detailed structure of VGG-16.

1) VGG-16: We first adopt the VGG-16 as the backbone network to encode the semantic information. The VGG-16 has been widely applied in previous aerial image semantic segmentation networks [34], [37], which can provide a fair comparison with our method. The detailed structure of the VGG-16 is shown in Fig. 3. We can see that it includes multiple convolution blocks and max-pooling layers. In detail, the convolution block consists of multiple convolution layers whose output channels are the same, and the output channel numbers of these blocks increase gradually accompanying with the increase of the network depth. Besides, the strides of the max-pooling layers are all set to 2, which can reduce the computation burden and increase the receptive field of the network significantly. In this article, we adopt the output feature maps of the third (stage-1), fourth (stage-2), and fifth (stage-3) max-pooling layers as the inputs of the information refinement block. In particular, the output resolutions of these feature maps are 1/8, 1/16, and 1/32 of the input images, respectively, which can ensure that the semantic information can be captured from three different scales as the same as that in [16] and [21]. In this article, we focus on the refinement of feature maps from stage-1 and stage-2 since these feature maps have large resolutions and can be used to encode more accurate small-scale semantic information from the aerial images.

2) ResNet-18: The second backbone network adopted in this work is the ResNet-18 and its detailed structure is shown in Fig. 4. The ResNet-18 first uses a convolution layer and a max-pooling layer to reduce the spatial resolution of the feature maps and then utilizes four blocks to encode more accurate semantic information. In particular, a block contains several units, and the channel numbers and resolution of a block can be controlled by its first unit. Specifically, the input resolution of a block is two times of its output resolution, as shown in Fig. 4(b). Besides, block1 possesses the same input and output resolution, and its detailed network is shown in Fig. 4(c). In this article, the output feature maps of the second (stage-1), third (stage-2), and fourth (stage-3) blocks are adopted as the inputs of the multiscale semantic information extractor. Similar to VGG-16, the output resolutions of these three blocks are 1/8, 1/16, and 1/32 of the input images, respectively. Compared with VGG-16, the convolution layers of ResNet-18 have less output channels, leading to a faster network.

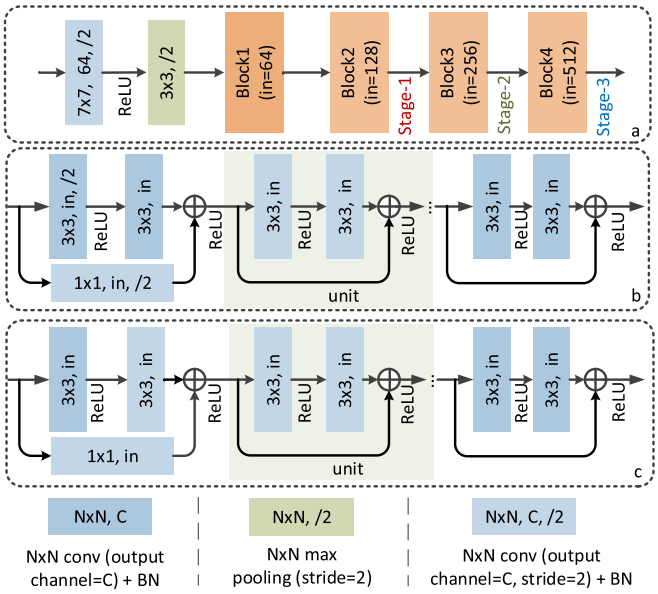

Fig. 4. Detailed structure of ResNet-18. (a) Whole network of ResNet-18. (b) Block whose input resolution is two times of its output resolution. (c) Block1 with the same resolution for input and output.

B. Information Refinement

Due to the shallow structure and limited information capturing ability of the backbone networks, directly fusing the feature maps generated by them to capture the multiscale semantic information can lead to an inferior segmentation result, especially for the small-scale object segmentation. Therefore, before the fusion of these feature maps, we propose two blocks to perform information refinement. In particular, we propose a channel attention refinement block to select informative features from the feature maps and a cross fusion block to increase the receptive field of the low-level feature maps. Next, we introduce these two blocks in detail.

1) Channel Attention Refinement Block: After acquiring multiple feature maps from the backbone networks, we propose a channel attention refinement block to refine them and improve their semantic information capturing ability. It is based on a popular deep learning technology, i.e., channel attention mechanism. Here, we introduce the channel attention mechanism and detail its specific implements in our proposed channel attention refinement block.

Channel attention has been used to select informative features via introducing global weights for different channels of a feature map [38]. Supposing that feature map $F \in \mathbb{R}^{C\times H\times W}$ is used as the input, $M_c \in \mathbb{R}^{C\times 1\times 1}$ is a global weight vector, and the output feature map F_{cout} refined by the channel attention can be calculated by

$$F_{\text{cout}} = F \otimes M_c \tag{1}$$

where $\otimes$ refers to the elementwise multiplication. With the multiplication, the global weights are broadcasted (i.e., copied) along the spatial dimension accordingly.

Generating a global weight vector is the most important process when applying the channel attention mechanism. In this article, motivated by [27], we generate the channel attention mask via a self-attention function and propose a

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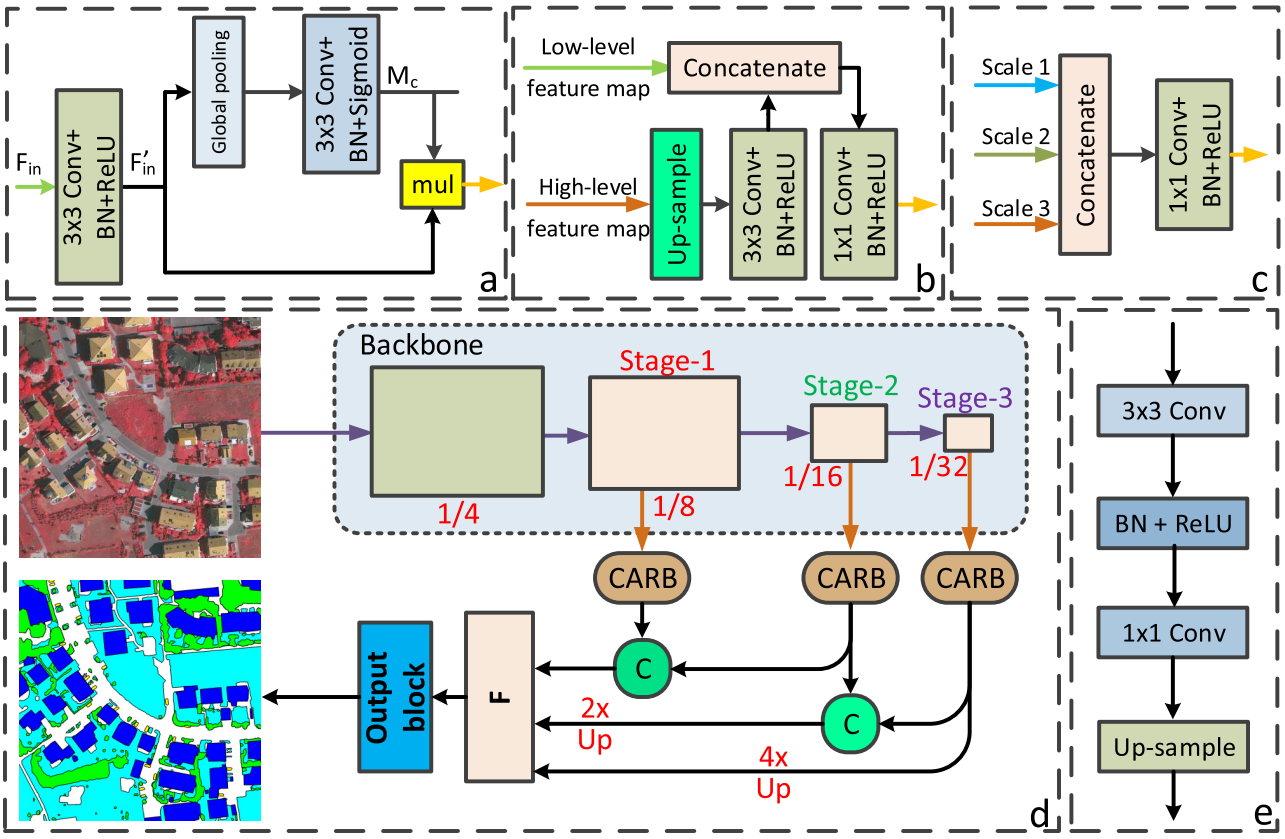

Fig. 5. Detailed blocks and whole network of the CF-Net. (a) Channel attention refinement block used to select informative features from the feature maps. (b) Cross fusion block used to increase the receptive field of low-level feature maps. (c) Information fusion block used to combine feature maps from different scales. (d) Network consisting of multiple blocks (CARB: channel attention refinement block; C: cross fusion block; F: information fusion block; $N\times$ UP: N times interpolation; and $1/N$: the resolution of input image is N times of this block). (e) Output block used to generate a feature map whose output channel number equals the class number.

channel attention refinement block, as shown in Fig. 5(a). In detail, we first use a 3×3 convolution with batch normalization (BN) and rectified linear unit (ReLU) to control the channel number of input feature map F_{in} and generate a new feature map F'_{in}. This convolution operation has two advantages. On the one hand, it can reduce the channel number of the input feature maps, which significantly reduces the computation burden of the network. On the other hand, it can unify the channel numbers of different level feature maps, which benefits the further fusion of them. Subsequently, we reduce the size of F'_{in} to 1×1 by the global average pooling operation and utilize 1×1 convolution with BN and sigmoid to generate the global weight vector M_c. Finally, we multiply M_c with F'_{in} to generate the output of the channel attention refinement block. With the channel attention refinement block, the network can capture more accurate semantic information from every scale.

2) Cross Fusion Block: Although the channel attention refinement block can enlarge the information capturing ability of the feature maps, the VGG-16 and ResNet-18 are two pretty shallow networks and their low-level feature maps still have a pretty small receptive field. Therefore, the feature maps used to capture small-scale objects still have inferior information capturing ability, which will further damage the segmentation results on the aerial images significantly. Motivated by [20], in this article, we propose a cross fusion block to address this problem. In detail, it can use the high-level feature maps to compensate for the receptive field of the low-level ones, and its detailed structure is shown in Fig. 5(b). The cross fusion block has two input feature maps F_{low} and F_{high}. In particular, the resolution of the high-level feature map F_{high} is $1/2$ of that of the low-level one, which can limit the information gap between these two feature maps and benefit the aggregation of them. We first use a bilinear interpolation operation UP$(\cdot)$ to change the resolution of the high-level feature map to the same as the low-level one and then use a 3×3 convolution layer with BN and ReLU (i.e., $\text{CV}_{3\times3}(\cdot)$) to refine the upsampled high-level feature map. Finally, we use a concatenation operation $\mathbb{C}(\cdot,\cdot)$ to concatenate the refined high-level feature map with the low-level one and adopt a 1×1 convolution layer with BN and ReLU (i.e., $\text{CV}_{1\times1}(\cdot)$) to generate the output F_{cfout} of the cross fusion block. The whole process can be denoted by

$$F_{\text{cfout}} = CV_{1\times1}(\mathbb{C}(F_{\text{low}}, CV_{3\times3}(Up(F_{\text{high}})))). \tag{2}$$

With the cross fusion block, the low-level feature maps will have similar receptive as the high-level ones, and their semantic information capturing ability can be then significantly improved. Besides, the cross fusion block has a pretty simple structure and the computation burden brought by it is small, leading to an efficient characteristic to capture more accurate small-scale semantic information.

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C. Information Fusion

The refinement results of the different level feature maps are finally fed into the information fusion block, whose structure is shown in Fig. 5(c). In detail, we first use concatenation operation to fuse the feature maps from different scales and then adopt a convolution layer with BN and ReLU to refine the fusion result. As a result, we can get a fused feature map that can capture semantic information from multiple scales.

D. Network Structure

The whole network used to combine the abovementioned blocks is shown in Fig. 5(d). In particular, we first use the backbone network to encode the semantic information of the aerial images and then fed the output feature maps of different stages into the channel attention refinement block to increase their information capturing ability. Subsequently, the refined feature maps from the first and second stages are fed into a cross fusion block, and the refined feature maps from the second and third stages are fed into another cross fusion block, which can increase the receptive field of the low-level feature maps significantly. After the refinement of the channel attention refinement block and cross fusion block, we upsample all the feature maps to the same resolution of the stage-1 and use the upsampled feature maps as the input of the information fusion block. Finally, we use an output block to process the output of the information fusion block and generate an output feature map whose resolution is the same as the input image and the number of output channels equals the class number of the data set. With these blocks, more accurate semantic information from different scales, especially from the small scale, can be extracted.

IV. Experimental Results

To demonstrate the superiority of the proposed CF-Net in the aerial scenes, two popular aerial image data sets are used in our experiments, i.e., ISPRS Vaihingen and Potsdam data sets. We first use the Vaihingen data set to conduct several ablation studies and then compare the CF-Net with other state-of-the-art networks on the Vaihingen and Potsdam data sets.

A. Data Sets and Training Details

1) Data Sets: The Vaihingen data set[1] is gathered from a 1.38 km^2 area of the city, Vaihingen. It consists of 33 aerial images with a spatial resolution of 9 cm. These aerial images have three bands (i.e., near-infrared (NIR), red (R), and green (G) wavelengths) and the average size of these images is 2494×2064. Moreover, the digital surface model (DSM) used to measure the distance between the object surfaces and the camera is included in the data sets. In this article, 16 images annotated by the pixelwise labels are selected as the training and testing data sets as the same as in [34], [39], and [40]. Among them, 11 images are used as the training set, and the other images (image IDs: 11, 15, 28, 30, and 34) are adopted to evaluate the CF-Net. In particular, the DSMs are not adopted in our experiments.

The Potsdam data set[2] is collected from a 3.42 km^2 area of the city Potsdam. It is composed of 33 aerial images with a spatial resolution of 5 cm. The resolution of these aerial images is 6000×6000, and these images have four channels [i.e., NIR, R, G, and blue (B)]. In our experiments, we select ten images to build the training set and seven images (image IDs: 02_11, 02_12, 04_10, 05_11, 06_07, 07_08, and 07_10) to construct the testing set, which obeys the setting of [34] and [39]. Specifically, we just use the R, G, and B channels in our experiments.

2) Training Details: To train our network and following [41], we set the initial learning rate $\mathrm{LR}_{\mathrm{ini}}$ to 0.01 and use a poly policy to reduce the learning rate (LR) gradually. The LR can be computed by

$$\mathrm{LR} = \mathrm{LR}_{\mathrm{ini}} \times \left(1 - \frac{\mathrm{iter}}{\mathrm{max_iter}}\right)^{\mathrm{power}}, \ \mathrm{power} = 0.9. \quad (3)$$

In this article, the max iteration number is set to 150 000. During the training stage, we perform several data augmentation strategies (i.e., randomly scaling images from 0.5 to 2 and flipping images) on the input images to improve the generalization ability of the network. We use a stochastic gradient descent (SGD) optimizer to train our network and set the batch size to 6, momentum to 0.9, and weight decay to 0.0005. In particular, all the backbones we adopted are pretrained on the ImageNet data set [42] because a good initialization can contribute a lot to the performance of the network [43], [44].

3) Loss Function: In our experiments, the training loss can be calculated by a pixelwise soft-max over the output feature map F combined with the cross-entropy loss function. The definition of the soft-max $p_k(X; F)$ is

$$p_k(X; F) = \frac{\exp(a_k(X; F))}{\sum_{i=1}^{K} \exp(a_i(X; F))} \quad (4)$$

where $a_k(X; F)$ denotes the activation of channel k on the output result F at the pixel position $X = (x, y)$ and K is the number of channels/classes. Specifically, $p_k(X; F)$ is the approximated maximum function, i.e., $p_k(X; F) \approx 1$ for the k with the maximum activation $a_k(X; F)$ and $p_k(X; F) \approx 0$ for all other k. The cross entropy $E(X; F, \mathrm{OH})$ then penalizes at position X, and it can be calculated as follows:

$$E(X; F, \mathrm{OH}) = -\sum_{k=1}^{K} \mathrm{OH}_k(X) \log(p_k(X; F)) \quad (5)$$

where $\mathrm{OH}(X)$ is the one-hot vector generated from the true label of the pixel at X. Then, the loss L of the output feature map F in the current training stage can be computed by

$$L_F = \frac{\sum_{b=1}^{B} \sum_{x=1}^{h} \sum_{y=1}^{w} E_b(X; F, \mathrm{OH})}{B \times h \times w} \quad (6)$$

where E_b is the cross entropy on the bth batch and B, h, and w represent the batch size, the height, and the width of the current training feature map, respectively.

[1] http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

[2] http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-potsdam.html

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TABLE I

ILLUSTRATING THE EFFECT OF EVERY BLOCK OF THE CF-NET. CARB MEANS THE CHANNEL ATTENTION REFINEMENT BLOCK AND CFB REPRESENTS THE CROSS FUSION BLOCK

Methods	CARB	CFB	mean F_1 (%)
CF-Net(VGG-16)	×	×	80.31
	✓	×	88.91
	×	✓	87.23
	✓	✓	89.97
CF-Net(ResNet-18)	×	×	79.69
	✓	×	88.02
	×	✓	86.92
	✓	✓	88.94

B. Ablation Experiments

In our ablation experiments, we train and evaluate our network on the Vaihingen set. The segmentation performance is evaluated by the F_1 score metric, which can be calculated as follows:

$$F_1 = (1+\beta^2)\cdot\frac{\text{precision}\cdot\text{recall}}{\beta^2\cdot\text{precision}+\text{recall}},\ \beta = 1. \tag{7}$$

Furthermore, the mean F_1 score can be calculated by averaging all F_1 scores of the foreground objects.

1) Baseline Network: To evaluate the performance gain that can be brought by the blocks we proposed, we first do not introduce any of them into the network and test the performance of the backbone networks. In particular, in the tail of the backbone networks, we add an output block [i.e., Fig. 5(e)] to construct a baseline network. We find that the mean F_1 scores of the VGG-16 and the ResNet-18 are 80.31% and 79.69%, respectively. Directly adopting the baseline networks can hardly meet the accuracy requirement of real-time applications. After obtaining this performance, we introduce all the blocks we proposed into the network and evaluate the performance. We find that the mean F_1 scores can improve significantly from 80.31% to 89.97% (VGG-16) and from 79.69% to 88.94% (ResNet-18), which proves the efficiency of our CF-Net. Specifically, the detailed performance gain brought by every block can be viewed in Table I.

2) Channel Attention Refinement Block: Attention mechanism refinement block is an efficient structure to enhance the information capturing ability of the network by selecting the informative features and suppressing the one of nonsense. Here, we perform several ablation studies to prove its advantage. In particular, we first remove the self-attention part from the channel attention refinement block. With this modification, the channel attention refinement block will degenerate into a normal convolution layer. We find that the mean F_1 scores decrease sharply from 89.97% to 87.23% (VGG-16) and from 88.94% to 86.92% (ResNet-18), which proves the importance of the self-attention part.

The second ablation experiment we conduct on the channel attention refinement block is the filter number of our CF-Net. From Fig. 5(a), we can see that a 3×3 convolution has been used to unify the channel number of the feature maps from

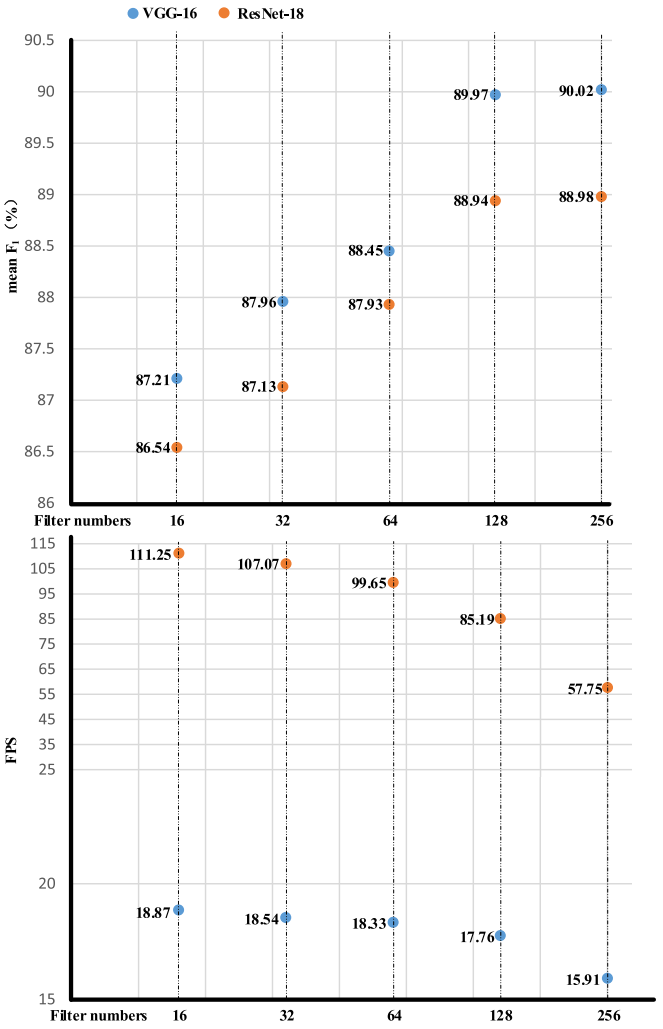

Fig. 6. Performance of network with different filter numbers in the channel attention refinement block.

different stages, and the output channel number of the feature maps from different stages can be regarded as the filter number of the CF-Net. Intuitively, more filters mean larger capacity to capture complex features, but increasing the filter numbers will increase the inference time of the network. At the same time, redundant information introduced by the unnecessary filters may hamper the network performance. To test its influence, we adopt the filter number at $\{16, 32, 64, 128, 256\}$, and the results are listed in Fig. 6. The inference speed is measured by the frames per second (FPSs) with the same size (i.e., $3 \times 1024 \times 1024$) of the input. The speed is tested on the PC equipped with an Intel Xeon CPU at 3.50 GHz and 16-GB RAM, together with an RTX 2080ti GPU. We can see that the increasing filter number will improve the accuracy of the network but requires more computational resources. Specifically, when the filter number is set to 128, the performance gain is small and the speed decrease is significant. Therefore, we set the filter number of the channel attention refinement block as 128, which can keep a good balance between accuracy and speed.

3) Cross Fusion Block: To measure the performance gain that can be brought by the cross fusion block, we remove

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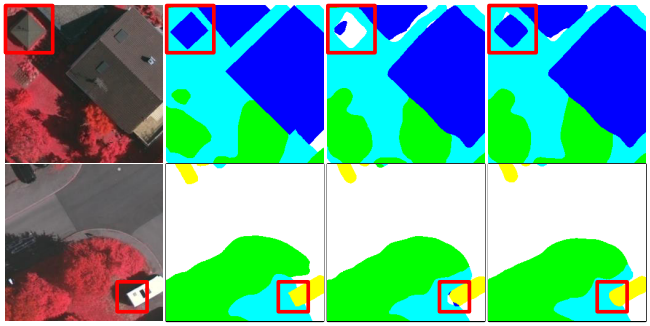

Fig. 7. Segmentation results with and without the cross fusion block (backbone VGG-16). The first column is the original images, the second column is the ground-truth labels, the third column is the segmentation results without cross fusion block, and the last column is the segmentation results with the cross fusion block.

this block from the network and test the performance change. Without the cross fusion block, the mean F_1 scores decrease sharply from 89.97% to 86.31% (VGG-16) and from 88.94% to 85.95% (ResNet-18), which proves the importance of the cross fusion block. Besides, we find that the F_1 score of the car suffers a more obvious decrease [i.e., from 91.35% to 85.12% (VGG-16) and 89.14% to 84.23% (ResNet-18)], which proves the importance of the cross fusion block for the small-scale semantic information capturing. Some visualization results are shown in Fig. 7. We see that the cross fusion block can result in better segmentation results on the small-scale objects. This is mainly due to the increase of the receptive field of the low-level feature maps. In particular, the high-level feature maps can be used to enlarge the receptive field of the low-level feature maps as illustrated in [20]. In the semantic segmentation field, the receptive field is pretty important for the semantic information capturing ability of the feature map [45]. Therefore, using cross fusion block can increase the information capturing ability of the low-level feature maps significantly, which can further improve the segmentation accuracy of the small-scale objects. Besides, we try to use the feature map from stage-3 as the high-level feature maps of the two cross fusion blocks, which can further increase the receptive field of the feature map from stage-1. However, we find that this modification will lead to performance degradation and the mean F_1 scores will decrease from 89.97% to 88.91% (VGG-16) and from 88.94% to 88.02% (ResNet-18). This may be caused by the information gap between the high- and low-level feature maps [20]. To remit this information gap, using the output feature map from stage-2 to guide that from stage-1 is more suitable.

4) Small-Scale Semantic Information Capturing: As discussed previously, the aerial images contain a large number of small-scale objects and the small-scale semantic information is of great importance for the segmentation of aerial scenes. In this section, we design an ablation experiment to verify this point. In particular, we focus on the F_1 score change of the cars due to that almost all of the cars are small-scale objects. Intuitively, the feature maps from low-level stages have larger resolution and can be used to extract semantic information from smaller scales. Therefore, we first remove the feature maps from the lowest stage (i.e., stage-1) and

TABLE II
CAPTURING SEMANTIC INFORMATION FROM DIFFERENT STAGES

Model Name	stages	Car (%)	mean F_1 (%)
CF-Net (VGG-16)	(3)	73.76	85.78
	(2, 3)	75.96	86.20
	(1, 2, 3)	91.35	89.97
CF-Net (ResNet-18)	(3)	72.45	84.93
	(2, 3)	74.89	85.55
	(1, 2, 3)	89.14	88.94

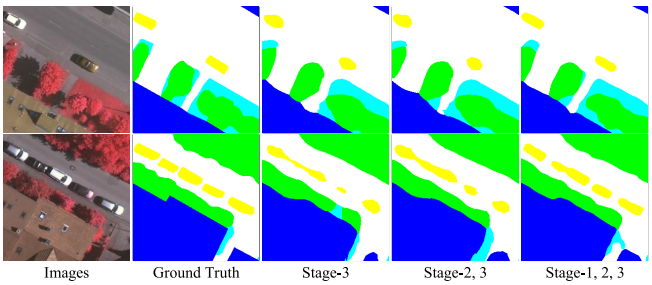

Fig. 8. Segmentation results when adopting feature maps from different stages as the inputs (baseline VGG-16).

do not capture semantic information from the small scale. Subsequently, we remove the feature maps from stage-1 and stage-2 simultaneously and just extract semantic information from the large scale. The results are shown in Table II. Clearly, F_1 scores of the cars decrease more significantly than the mean F_1 scores. This is because semantic information of the cars can hardly be extracted from a large scale. Some visualization results are shown in Fig. 8.

C. Comparison With State of the Art

To prove the superiority of our CF-Net, in this section, we compare it with several state-of-the-art semantic segmentation approaches on the Vaihingen and Potsdam data set benchmarks. Specifically, the mean IoU (mIoU) and overall accuracy (OA) are also adopted in this section for a comprehensive comparison. The mIoU metric can be computed by

$$\text{mIoU} = \frac{1}{k}\sum_{i=1}^{k}\frac{\text{TP}_i}{\text{FN}_i + \text{FP}_i + \text{TP}_i} \tag{8}$$

where FN_i, FP_i, and TP_i represent the false negative, false positive, and true positive pixel numbers of the ith class, respectively, and k denotes the number of classes in the data set. The OA can be calculated as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{N} \tag{9}$$

where TP, TN, and N are the numbers of true positive and true negative, and the total pixel number of the whole image, respectively.

1) Comparison on the Vaihingen Data Set: We first compare our CF-Net with multiple state-of-the-art methods on the Vaihingen data set from two aspects: accuracy and inference speed. We choose eight representative state-of-the-art aerial image semantic segmentation networks as the comparing

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TABLE III

COMPARISON WITH STATE-OF-THE-ART AERIAL IMAGE SEMANTIC SEGMENTATION METHODS ON THE VAIHINGEN DATA SET. BOLD INDICATES THE BEST

Model Name	Imp. surf.	Build.	Low veg.	Tree	Car	mean F_1 (%)	mIoU (%)	OA (%)	FPS
FCN [16]	88.67	92.83	76.32	86.67	74.21	83.74	72.69	86.51	16.54
SCNN [46]	88.21	91.80	77.17	87.23	78.60	84.40	73.73	86.43	9.11
PSPNet (VGG-16) [19]	89.92	94.36	78.19	87.12	72.97	84.51	73.97	87.62	5.54
SegNet+Distance maps [47]	91.47	94.76	81.91	88.49	74.01	96.12	-	89.2	-
RotEqNet [48]	89.50	94.80	77.50	86.50	72.60	84.18	-	87.50	-
UFMG_4 [49]	91.10	94.50	82.90	88.80	81.30	87.72	-	89.40	6.32
S-RA-FCN [34]	91.47	94.97	80.63	88.57	87.05	88.54	79.76	89.23	-
HMANet (VGG-16) [41]	91.86	94.52	**83.17**	**89.81**	87.15	89.30	80.68	89.95	4.87
CF-Net (ResNet-18)	91.43	95.07	80.29	88.79	89.14	88.94	80.43	89.28	**85.19**
CF-Net (VGG-16)	**92.30**	**95.58**	81.32	89.29	**91.35**	**89.97**	**82.10**	**89.99**	17.76

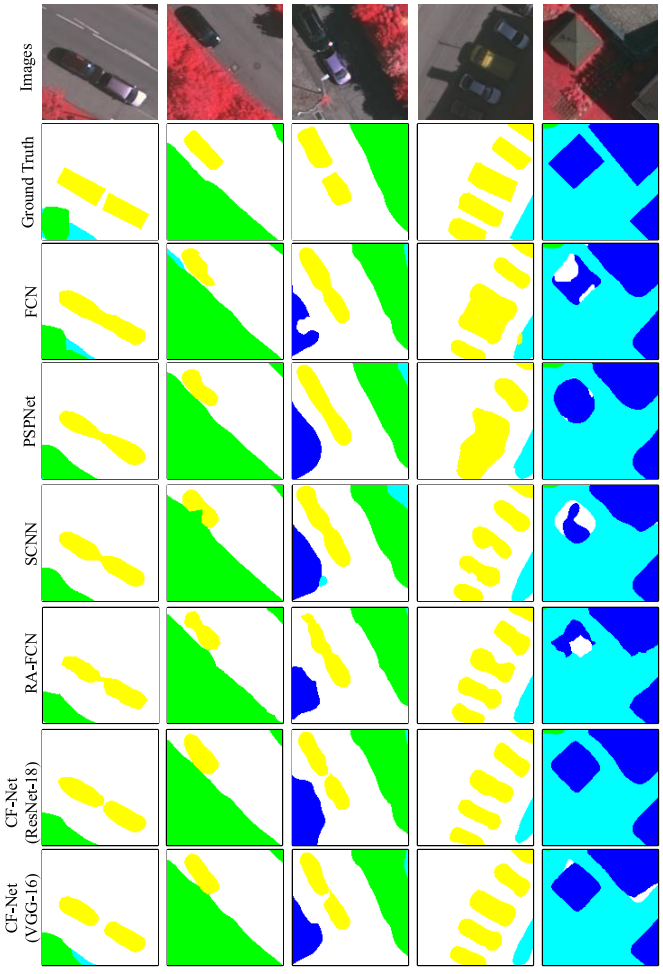

Fig. 9. Segmentation results (cropped images) on Vaihingen. Legend—white: impervious surfaces, blue: buildings, cyan: low vegetation, green: trees, yellow: cars, and red: clutter/background.

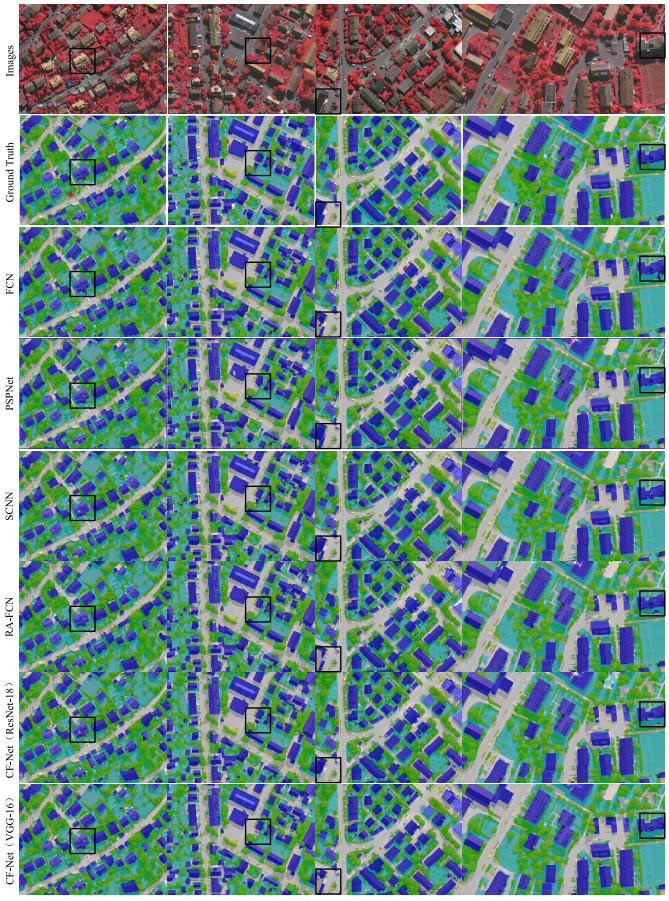

Fig. 10. Segmentation results (whole images) on Vaihingen. Several large-scale objects are marked by the black boxes.

methods. To compare the inference speed, we measure it by the FPSs with the same size (i.e., $3 \times 1024 \times 1024$) of input. In this article, all methods are tested on the same PC equipped with an Intel Xeon CPU at 3.50 GHz and 16-GB RAM, together with an RTX 2080ti GPU. The results are shown in Table III. Compared with other state-of-the-art methods, our method can achieve the highest accuracy with the fastest speed. This is mainly due to the proposed efficient refinement blocks. On the one hand, they can enhance the information capturing ability of the network significantly, leading to superior segmentation results. On the other hand, they can prevent the introduction of complex structures used for the further refinement and fusion of different level feature maps, which contribute a lot to the fast speed of our CF-Net. Moreover, we notice that our method can surpass other methods on the small-scale objects

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TABLE IV
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE POTSDAM DATA SETS. BOLD INDICATES THE BEST

Model Name	Imp. surf.	Build.	Low veg.	Tree	Car	mean F_1	mIoU (%)	OA	FPS
FCN [16]	88.67	92.83	76.32	79.83	74.21	83.74	78.34	85.59	16.54
SCNN [46]	88.37	92.32	83.68	80.94	91.17	84.22	77.72	85.57	9.11
Dilated FCN [24]	86.52	90.78	83.01	78.41	90.42	85.57	-	84.14	-
UZ_1 [40]	89.30	95.40	81.80	80.50	86.50	86.70	-	85.80	-
PSPNet (VGG-16) [19]	86.99	93.08	85.55	83.81	93.22	88.58	79.64	85.94	5.54
UFMG_4 [49]	90.80	**95.60**	84.40	84.30	92.40	89.50	-	87.90	6.32
S-RA-FCN [34]	**91.33**	94.70	**86.81**	83.47	94.52	90.17	82.38	**88.59**	-
CF-Net (ResNet-18)	90.95	93.19	86.19	84.49	**95.53**	90.07	82.20	88.03	**85.19**
CF-Net (VGG-16)	90.88	94.18	86.51	**84.73**	**95.53**	**90.37**	**82.69**	88.34	17.76

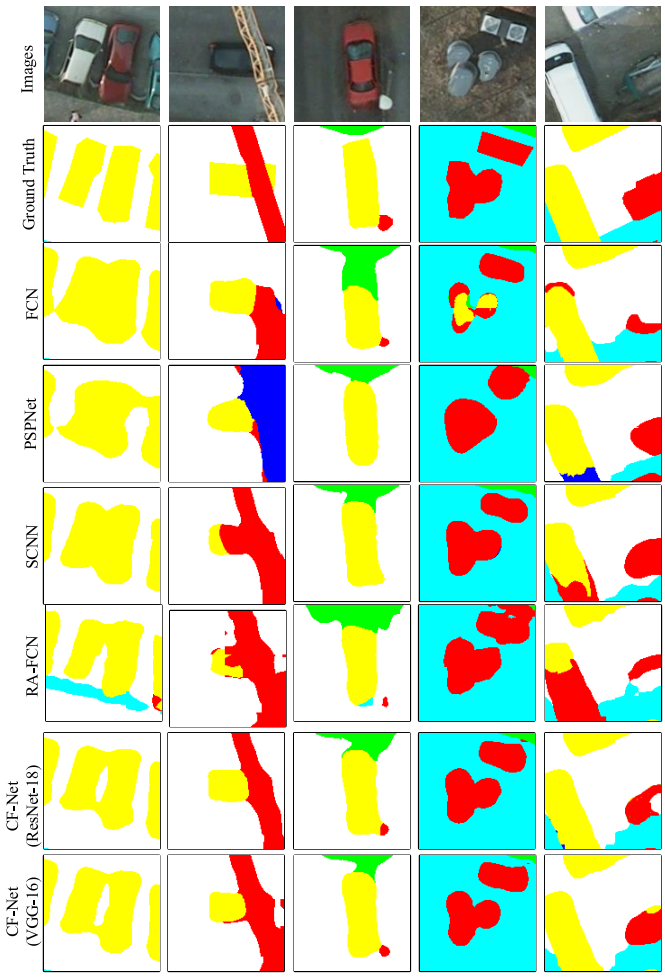

Fig. 11. Segmentation results (cropped images) on Potdsam.

with a large margin (i.e., the F_1 score improvement on the cars is about 4.20% and the average F_1 score improvement is about 0.47%), which proves that our method is more suitable for the small-scale semantic information capturing in aerial scenes. Some visual results on the Vaihingen data set are shown in Figs. 9 and 10. In particular, Fig. 9 shows some segmentation results of the image patches whose resolutions are 128×128, and Fig. 10 shows the segmentation results of the whole images. From the cropped images, we can see that our method has a more powerful ability to handle small-scale objects. However, compared with other methods, our method does not have obvious advantages to extract the large-scale semantic information as illustrated in the black boxes of the whole images, which is the main problem we should address in our future work.

TABLE V
COMPARING THE CF-NET WITH STATE-OF-THE-ART AERIAL IMAGE SEMANTIC SEGMENTATION METHODS ON THE HIGH-QUALITY SEMANTIC SEGMENTATION TASK. THE EXPERIMENTS ARE IMPLEMENTED ON THE VAIHINGEN DATA SET. BOLD INDICATES THE BEST

Model Name	backbone network	mean F_1 (%)
CAM-DFCN [50]	ResNet-101	87.11
CASIA2 [51]	ResNet-101	90.10
CCNet [52]	ResNet-101	90.58
CF-Net	ResNet-101	**90.73**

2) Comparison on the Potsdam Data Set: To further verify the superiority of our CF-Net, we perform experiments on the Potsdam data set, and the results are reported in Table IV. Clearly, our method again has both the highest accuracy and the fastest inference speed. We can see that compared with the Vaihingen data set, the performance gain brought by the CF-Net on the Potsdam data set is smaller. This may be caused by two reasons. On the one hand, the Potsdam data set has a larger image resolution under the same FOV. Hence, it contains fewer small-scale objects, leading to the performance decrease of our method. On the other hand, we do not adopt the DSMs and NIR channel to train our CF-Net, which further limits the segmentation performance of our CF-Net. Some visual results on the Potsdam data set are shown in Figs. 11 and 12. Fig. 11 shows some segmentation results of the image patches whose resolutions are 128×128, and Fig. 12 shows the segmentation results of the whole images.

3) High-Quality Segmentation: To evaluate the performance of the CF-Net on the high-quality segmentation task, we replace the backbone network of it from VGG-16 to ResNet-101 and compare it with several methods whose backbone networks are complex. The experimental results on

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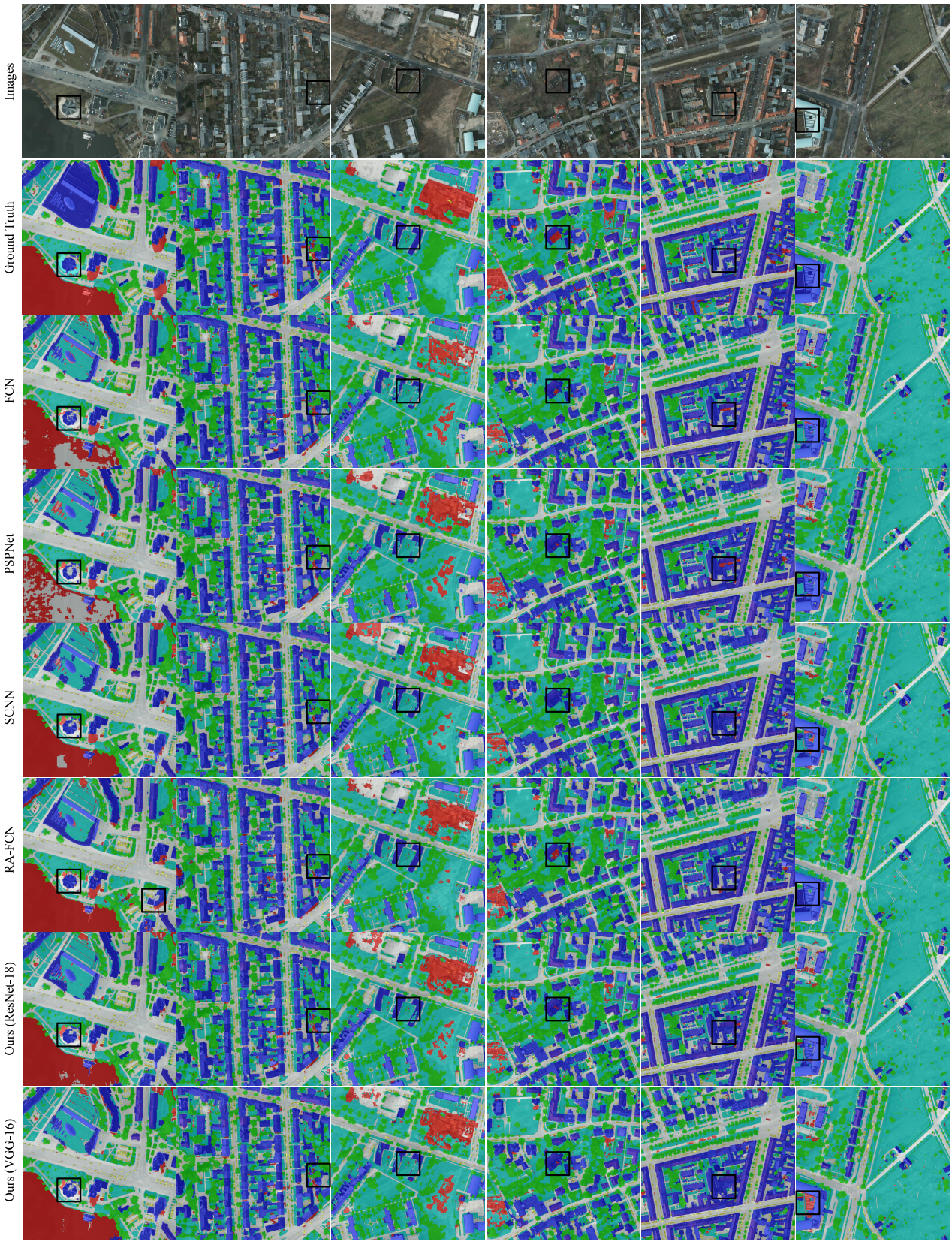

Fig. 12. Segmentation results (whole images) on Potsdam. Several large-scale objects are marked by the black boxes.

the Vaihingen data set are shown in Table V. We can see that the proposed CF-Net also has the best accuracy on the high-quality segmentation task.

V. Conclusion

This article develops a fast semantic segmentation network called CF-Net for multiscale semantic information capturing,

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especially for the small-scale semantic information capturing in aerial scenes. Our CF-Net can refine the hierarchical feature maps in two ways. On the one hand, the CF-Net can select informative features by the channel attention mechanism. On the other hand, the CF-Net can enlarge the receptive field of the low-level feature maps by inserting more abstract information into them. Therefore, it can capture more accurate semantic information from different scales. Moreover, our CF-Net has a simple yet efficient structure, leading to its fast inference speed. We verify our method on the Potsdam and Vaihingen data sets. Compared with other state-of-the-art approaches, our method can attain higher accuracy with less inference time, which proves its superiority. In the future, we will pay more attention to improve the segmentation accuracy of the large-scale objects and further improve the inference speed of the network.

REFERENCES

[1] J. C. Tilton, W. T. Lawrence, and A. J. Plaza, "Utilizing hierarchical segmentation to generate water and snow masks to facilitate monitoring change with remotely sensed image data," *GIScience Remote Sens.*, vol. 43, no. 1, pp. 39–66, Mar. 2006.

[2] L. Lopez-Fuentes, C. Rossi, and H. Skinnemoen, "River segmentation for flood monitoring," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2017, pp. 3746–3749.

[3] H. Sheng, X. Chen, J. Su, R. Rajagopal, and A. Ng, "Effective data fusion with generalized vegetation index: Evidence from land cover segmentation in agriculture," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 60–61.

[4] J. K. Jadhav, A. P. Sonavale, and R. Singh, "Segmentation analysis using particle swarm optimization-self organizing map algorithm and classification of remote sensing data for agriculture," in *Proc. Int. Conf. Intell. Data Commun. Technol. Internet Things*, 2019, pp. 659–668.

[5] S. Chen, T. Sun, F. Yang, H. Sun, and Y. Guan, "An improved optimum-path forest clustering algorithm for remote sensing image segmentation," *Comput. Geosci.*, vol. 112, pp. 38–46, Mar. 2018.

[6] L. Mi and Z. Chen, "Superpixel-enhanced deep neural forest for remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 140–152, Jan. 2020.

[7] M. M. Nielsen, "Remote sensing for urban planning and management: The use of window-independent context segmentation to extract urban features in Stockholm," *Comput., Environ. Urban Syst.*, vol. 52, pp. 1–9, Jul. 2015.

[8] Y. Liu, L. Gross, Z. Li, X. Li, X. Fan, and W. Qi, "Automatic building extraction on high-resolution remote sensing imagery using deep convolutional encoder-decoder with spatial pyramid pooling," *IEEE Access*, vol. 7, pp. 128774–128786, 2019.

[9] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 2097–2104.

[10] A. Radman, N. Zainal, and S. A. Suandi, "Automated segmentation of iris images acquired in an unconstrained environment using HOG-SVM and GrowCut," *Digit. Signal Process.*, vol. 64, pp. 60–70, May 2017.

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.

[12] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[13] N. He, L. Fang, S. Li, J. Plaza, and A. Plaza, "Skip-connected covariance network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1461–1474, May 2020.

[14] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[15] C. Yu, Y. Liu, C. Gao, C. Shen, and N. Sang, "Representative graph neural network," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 379–396.

[16] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[17] C. Yu, J. Wang, C. Gao, G. Yu, C. Shen, and N. Sang, "Context prior for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12416–12425.

[18] N. He, L. Fang, and A. Plaza, "Hybrid first and second order attention Unet for building segmentation in remote sensing images," *Sci. China Inf. Sci.*, vol. 63, no. 4, Apr. 2020, Art. no. 140305.

[19] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[20] C. Peng and J. Ma, "Semantic segmentation using stride spatial pyramid pooling and dual attention decoder," *Pattern Recognit.*, vol. 107, Nov. 2020, Art. no. 107498.

[21] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*. [Online]. Available: http://arxiv.org/abs/1706.05587

[22] F. Yu, V. Koltun, and T. Funkhouser, "Dilated residual networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 636–644.

[23] M. Yang, K. Yu, C. Zhang, Z. Li, and K. Yang, "DenseASPP for semantic segmentation in street scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3684–3692.

[24] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[25] H. Zhao, X. Qi, X. Shen, J. Shi, and J. Jia, "Icnet for real-time semantic segmentation on high-resolution images," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 405–420.

[26] Z. Wu, C. Shen, and A. van den Hengel, "Real-time semantic image segmentation via spatial sparsity," 2017, *arXiv:1712.00213*. [Online]. Available: http://arxiv.org/abs/1712.00213

[27] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "Bisenet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 325–341.

[28] R. P. Poudel, S. Liwicki, and R. Cipolla, "Fast-SCNN: Fast semantic segmentation network," in *Proc. Brit. Mach. Vis. Conf.*, 2019, pp. 1–12.

[29] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[30] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for scene segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[31] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5168–5177.

[32] V. Nekrasov, C. Shen, and I. Reid, "Light-weight refinenet for real-time semantic segmentation," in *Proc. Brit. Mach. Vis. Conf.*, 2018, pp. 1–15.

[33] S. Liu, W. Ding, C. Liu, Y. Liu, Y. Wang, and H. Li, "ERN: Edge loss reinforced semantic segmentation network for remote sensing images," *Remote Sens.*, vol. 10, no. 9, p. 1339, Aug. 2018.

[34] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[35] F. Zhou, R. Hang, and Q. Liu, "Class-guided feature decoupling network for airborne image segmentation," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 15, 2020, doi: 10.1109/TGRS.2020.3006872.

[36] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*. [Online]. Available: http://arxiv.org/abs/1409.1556

[37] J. Sherrah, "Fully convolutional networks for dense semantic labelling of high-resolution aerial imagery," 2016, *arXiv:1606.02585*. [Online]. Available: http://arxiv.org/abs/1606.02585

[38] M. Zhu, J. Li, N. Wang, and X. Gao, "A deep collaborative framework for face photo–sketch synthesis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 10, pp. 3096–3108, Oct. 2019.

[39] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "High-resolution aerial image labeling with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7092–7103, Dec. 2017.

[40] M. Volpi and D. Tuia, "Dense semantic labeling of subdecimeter resolution images with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 881–893, Feb. 2017.

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[41] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," 2020, *arXiv:2001.02870*. [Online]. Available: http://arxiv.org/abs/2001.02870

[42] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[43] J. Xie, N. He, L. Fang, and A. Plaza, "Scale-free convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6916–6928, Sep. 2019.

[44] M. Zhu, J. Li, N. Wang, and X. Gao, "Knowledge distillation for face photo-sketch synthesis," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Oct. 27, 2020, doi: 10.1109/TNNLS.2020.3030536.

[45] C. Peng, X. Zhang, G. Yu, G. Luo, and J. Sun, "Large kernel matters—Improve semantic segmentation by global convolutional network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4353–4361.

[46] X. Pan, J. Shi, P. Luo, X. Wang, and X. Tang, "Spatial as deep: Spatial CNN for traffic scene understanding," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 4278–4284.

[47] D. Chai, S. Newsam, and J. Huang, "Aerial image semantic segmentation using DCNN predicted distance maps," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 309–322, Mar. 2020.

[48] D. Marcos, M. Volpi, B. Kellenberger, and D. Tuia, "Land cover mapping at very high resolution with rotation equivariant CNNs: Towards small yet accurate models," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 96–107, Nov. 2018.

[49] K. Nogueira, M. D. Mura, J. Chanussot, W. R. Schwartz, and J. A. D. Santos, "Dynamic multicontext segmentation of remote sensing images based on convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7503–7520, Oct. 2019.

[50] H. Luo, C. Chen, L. Fang, X. Zhu, and L. Lu, "High-resolution aerial images semantic segmentation using deep fully convolutional network with channel attention mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3492–3507, Sep. 2019.

[51] Y. Liu, B. Fan, L. Wang, J. Bai, S. Xiang, and C. Pan, "Semantic labeling in very high resolution images via a self-cascaded convolutional neural network," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 78–95, Nov. 2018.

[52] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

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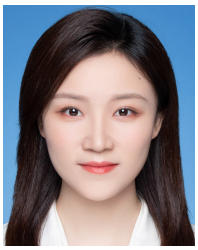

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